

then some v_j (with $j > 1$) is a linear combination of the preceding vectors, $v_1, \dots, v_{j-1}$. $\square$

Warning: Theorem 7 does not say that every vector in a linearly dependent set is a linear combination of the preceding vectors. A vector in a linearly dependent set may fail to be a linear combination of the other vectors.

Example 4: Let $u = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix}$. Describe the set spanned by u and v , and explain why a vector w is in $\text{Span}\{u, v\}$ if and only if $\{u, v, w\}$ is linearly dependent.

Solution: Let x_1 and x_2 be any two arbitrary real numbers.

$$x_1 \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}. \text{ We need to find out the possible values of } a, b \text{ and } c.$$

By defn. of the matrix equation $Ax=b$, we have: $\begin{bmatrix} 3 & 1 \\ 1 & 6 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

From Theorem 3, the matrix equation $Ax=b$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ b]$

$$\begin{bmatrix} 3 & 1 & a \\ 1 & 6 & b \\ 0 & 0 & c \end{bmatrix} \xrightarrow[\text{Row 2 and Row 1}]{\text{Interchange}} \begin{bmatrix} 1 & 6 & b \\ 3 & 1 & a \\ 0 & 0 & c \end{bmatrix} \xrightarrow[\text{-3xRow 1}]{\text{Row 2 = Row 2}} \begin{bmatrix} 1 & 6 & b \\ 0 & -17 & a-3b \\ 0 & 0 & c \end{bmatrix} \xrightarrow[\text{x-(1/17)}]{\text{Row 2 = Row 2}} \begin{bmatrix} 1 & 6 & b \\ 0 & 1 & (3b-a)/17 \\ 0 & 0 & c \end{bmatrix}$$

$$\begin{bmatrix} 1 & 6 & b \\ 0 & 1 & (3b-a)/17 \\ 0 & 0 & c \end{bmatrix} \xrightarrow[\text{-6xRow 2}]{\text{Row 1 = Row 1}} \begin{bmatrix} 1 & 0 & b - \frac{18b-6a}{17} \\ 0 & 1 & (3b-a)/17 \\ 0 & 0 & c \end{bmatrix}$$

$$\text{The corresponding system of equations are: } x_1 = b - \frac{18b-6a}{17} = \frac{6a-b}{17}$$

$$\text{and } x_2 = \frac{3b-a}{17} \text{ and } c=0.$$

It's easy to see that a and b can take any two arbitrary real values. $\therefore$ The set of vectors spanned are of the form $\begin{bmatrix} a \\ b \\ 0 \end{bmatrix}$.